

Breathing Training and Acute Breath-Regulation Strategies in Sport and Performance Contexts: A Scoping Review

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June 18, 2026

Abstract

Background: Breath-regulation practices are increasingly used in sport, exercise, and performance settings, but the evidence base spans heterogeneous interventions, populations, outcomes, and study designs. **Objective:** This scoping review mapped the available evidence on breathing-training and acute breath-regulation strategies in athletic or performance-relevant contexts, summarized study characteristics, charted intervention categories and outcomes, and identified evidence gaps for future trials and reviews. **Eligibility criteria:** Sources were eligible when they examined a breathing intervention or breathing-related exposure relevant to sport, exercise, or performance, including hyperventilation, hypoventilation, paced breathing, pranayama/yogic breathing, breath holding, hook breathing, Wim Hof-type breathing, or closely related approaches. **Sources of evidence:** The review started from 272 PubMed records, retained 43 after title/abstract screening, retained 31 after full-text screening, and recovered three additional papers by citation backtracking. **Charting and synthesis:** Data were charted into bibliometric, demographic, intervention, outcome, feasibility, and safety tables, then summarized descriptively following PRISMA-ScR guidance. Where sufficiently comparable outcome clusters existed, exploratory standardized mean differences were calculated. **Results:** Thirty-four sources were included. Most were experimental (31/34, 91.2%), but many were crossover, single-session, small-sample, or otherwise classified as experimental-other. Evidence was concentrated in hyperventilation, paced breathing, hypoventilation, pranayama/yogic breathing, and breath-holding strategies, with broad variation in sport context and outcome selection. Exploratory pooling suggested a small, borderline effect favoring hyperventilation for power outcomes ($g = 0.30$, 95% CI -0.00 to 0.60), while paced-breathing effects on reaction outcomes were inconclusive ($g = -0.15$, 95% CI -0.45 to 0.15). Feasibility reporting was limited; adherence and dropout were not parseable in the cleaned dataset. **Conclusions:** The field is active but fragmented. Current evidence supports cautious hypothesis generation rather than definitive practice recommendations. Future work should use preregistered protocols, clearer intervention taxonomies, sport-specific outcomes, adverse-event reporting, and adequately powered comparative designs.

Keywords: scoping review; PRISMA-ScR; breathing training; breathwork; hyperventilation; hypoventilation; paced breathing; sport performance; athletes

1 Introduction

Breathing is both an automatic physiological process and a deliberately modifiable behavior. In sport and exercise settings, athletes and practitioners use breath-regulation strategies to influence arousal, autonomic control, acid-base balance, tolerance to hypoxia or hypercapnia, respiratory muscle function, re-

covery, and perceived exertion. The practical appeal is clear: breathing techniques are low cost, portable, and easy to combine with warm-up, recovery, psychological preparation, or technical routines. The scientific challenge is equally clear: the term “breathing training” covers several mechanistically different practices that should not be assumed to have the same effects.

The interventions represented in the AIK-SCR extraction files range from acute hyperventilation or Wim Hof-type breathing to repeated-sprint hypoventilation, slow-paced or coherent breathing, box breathing, pranayama, Bhramari breathing, breath holding, Valsalva-type strategies, hook breathing in freediving, and observational comparisons of breathing patterns. These practices plausibly act through different pathways. Hyperventilation may acutely alter carbon dioxide, pH, ventilatory discomfort, and repeated high-intensity exercise tolerance. Hypoventilation or breath-holding strategies may expose athletes to hypoxemic or hypercapnic stress. Paced breathing is more often framed as an autonomic, attentional, or anxiety-regulation strategy. Pranayama and related yogic methods combine respiratory control with broader psychophysiological training.

This heterogeneity makes a scoping review more appropriate than a narrowly framed systematic review of efficacy. Scoping reviews are designed to map a body of literature, clarify concepts, identify types of available evidence, and locate gaps where later systematic reviews or trials may be justified (Arksey & O’Malley, 2005; Levac et al., 2010; Peters et al., 2020). Reporting followed the PRISMA extension for scoping reviews (PRISMA-ScR), which emphasizes transparent reporting of eligibility criteria, evidence sources, selection, data charting, and synthesis (Tricco et al., 2018).

The objective of this review was to map evidence on breathing-training and acute breath-regulation strategies in sport, exercise, and performance-relevant contexts. Specifically, we aimed to describe publication trends, study designs, populations, breathing-intervention categories, dosing characteristics, outcome domains, feasibility and safety reporting, and any comparable quantitative outcome clusters emerging from the extracted material.

2 Methods

2.1 Design and reporting framework

This manuscript is a scoping review reported according to PRISMA-ScR (Tricco et al., 2018). The review used the AIK-SCR project extraction files as the data source for manuscript synthesis. A formal protocol or registry record was not available in the project folder at the time of manuscript generation.

2.2 Eligibility criteria

Sources were considered eligible when they examined a breathing intervention, breathing maneuver, or breathing-related exposure relevant to sport, exercise, or performance. Eligible concepts included hyperventilation, hypoventilation, paced or coherent breathing, box breathing, pranayama or yogic breathing, breath holding, Valsalva-type strategies, hook breathing, Wim Hof-type breathing, diaphragmatic breathing, and similar practices. Eligible contexts included athletes, trained or recreationally active participants, sport-specific tasks, exercise performance, recovery, psychophysiological preparation, and laboratory performance tasks with plausible sport relevance. No formal restrictions on study design were applied during charting.

2.3 Information sources and selection

The review began with 272 records from PubMed. After title and abstract screening, 43 records were retained for full-text assessment. Thirty-one reports were retained after full-text screening. Citation backtracking recovered three additional reports: Taylor 2021, Fernandez 2019, and Fernandez 2025. The final extracted evidence base therefore contained 34 sources. The exact PubMed search string, search date, duplicate-removal log, and full-text exclusion reasons were not present in the current project files and should be inserted if recovered.

2.4 Data charting

Data were charted into six source tables: bibliography, descriptive characteristics, intervention specifics, intervention outcomes, control/comparator outcomes, and feasibility/safety. The analysis scripts cleaned author-year identifiers, parsed numeric variables, standardized breathing categories, classified broad study design groups, and exported cleaned tables and plots. Charted variables included publication year, study design, sport discipline, athlete level, sample size, age, sex distribution, breathing classification, dose frequency, dose duration, comparator type, outcome metric, group sample sizes, group means and dispersions, effect estimates, adverse events, adherence, dropout, and feasibility scores.

2.5 Synthesis

The primary synthesis was descriptive and evidence-mapping oriented. Publication year, study design, sport context, participant characteristics, breathing categories, dose frequency, dose duration, outcome clusters, and feasibility/safety fields were summarized from the cleaned outputs. For outcome clusters with at least two sufficiently comparable complete records, exploratory Hedges' g values and random-effects models were generated by the existing R scripts. These analyses were treated as exploratory adjuncts to the scoping review rather than as definitive meta-analytic evidence.

2.6 Critical appraisal

Consistent with PRISMA-ScR, formal risk-of-bias appraisal was not required for the central mapping objective. Instead, the review documents data completeness issues, missing identifiers, unparseable sample sizes, incomplete outcome fields, and limited feasibility reporting. These issues are interpreted as evidence-map limitations rather than exclusion criteria.

3 Results

3.1 Selection of sources

The selection process is shown in Figure 1. From 272 PubMed records, 43 were retained after title and abstract screening. Thirty-one were included after full-text screening, and three additional papers were recovered by citation backtracking, yielding 34 included sources.

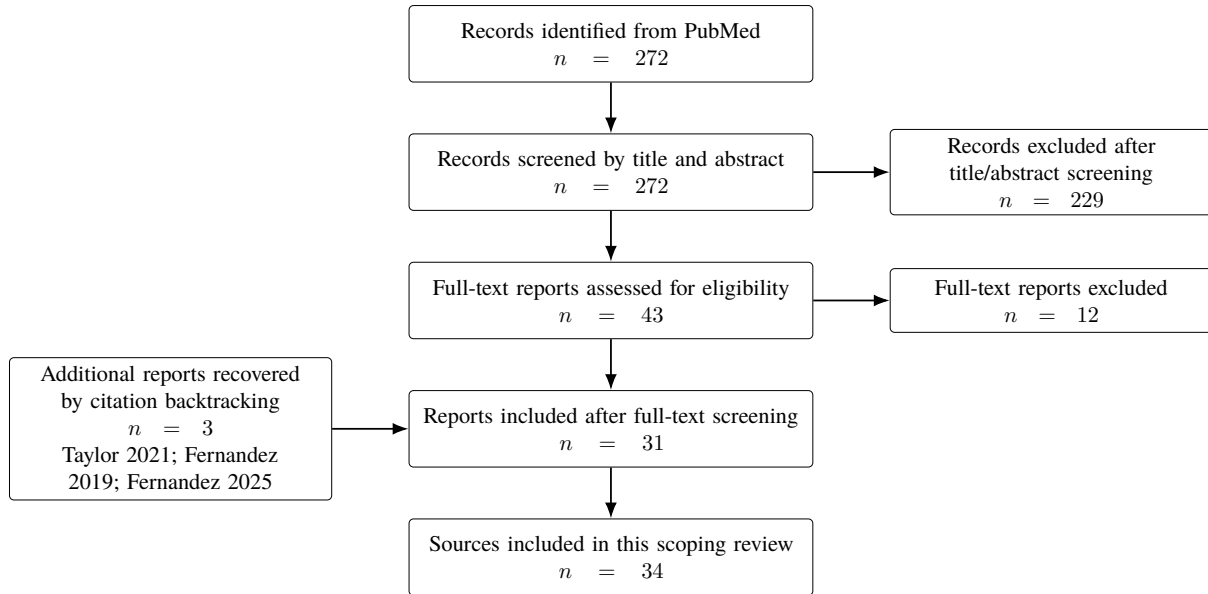


Figure 1: PRISMA-ScR source selection flow. Duplicate-removal counts and full-text exclusion reasons were not available in the project files and should be completed if those logs are recovered.

Table 1: Summary of the mapped evidence base.

Domain	Finding
Initial PubMed records	272
Title/abstract retained	43
Full-text retained	31
Citation-backtracked additions	3
Included sources in extraction dataset	34
Publication years represented	2004–2026 in cleaned extraction; one source has a 2000 bibliography year
Experimental studies	31/34 (91.2%)
Observational studies	2/34 (5.9%)
Unclear design classification	1/34 (2.9%)
Studies with age data	26 studies, 1185 participants
Weighted mean age	26.7 years (pooled SD 10.2)
Studies with sex data	1234 participants; estimated male proportion 51.5%

Table 2: Breathing-training categories in the extracted intervention records.

Category	Records	Percentage
Other or unclear	11	31.4%
Hyperventilation	9	25.7%
Paced breathing	9	25.7%
Hypoventilation	3	8.6%
Pranayama/yogic breathing	2	5.7%
Breath holding/Valsalva	1	2.9%

Table 3: Most represented sport contexts.

Sport context	Studies	Percentage
Not sport-specific	7	20.6%
Various/mixed sports	6	17.6%
Freediving	3	8.8%
Running	3	8.8%
Swimming	3	8.8%
Resistance training	2	5.9%
Single-study contexts	10	29.4%

Table 4: Exploratory random-effects syntheses from comparable outcome clusters.

Cluster	k	Pooled g	95% CI	p	I^2
Hyperventilation and power outcomes	6	0.30	-0.00 to 0.60	0.052	0.0%
Paced breathing and reaction outcomes	4	-0.15	-0.45 to 0.15	0.335	57.9%
Hypoventilation and repeated-sprint outcomes	1	–	Not pooled	–	–

Positive values were coded as favoring the breathing intervention after direction normalization. These syntheses are exploratory because outcome clusters were small and heterogeneous.

Table 5: Feasibility and safety reporting.

Item	Extracted finding
Studies with parseable adherence data	0
Studies with parseable dropout data	0
Adverse event mentions	Dizziness, heaviness, numbness, tingling, and pulmonary edema
Main safety pattern	Hyperventilation-related sensory symptoms were reported in one source; pulmonary edema was noted in a freediving/hook-breathing context

3.2 Characteristics of included sources

The cleaned dataset mapped 34 sources spanning 2004–2026, with one bibliographic discrepancy in which the extraction identifier ‘Durand2004’ corresponds to a 2000 BibTeX record. Publication volume increased in recent years, with the highest annual counts in 2025 and 2026 in the cleaned extraction table. Most studies were experimental (31/34, 91.2%), while two were observational and one remained unclear. However, only eight records were classified as parallel or controlled trials; most were crossover, repeated-measures, acute, pilot, or otherwise categorized as experimental-other.

Across studies with age data, 1185 participants contributed to a weighted mean age of 26.7 years. Sex data were available for 1234 participants and suggested an approximately balanced overall sample, with an estimated male proportion of 51.5%. Sport contexts were dispersed. The most frequent categories were not sport-specific, mixed/various sports, freediving, running, and swimming, while many sport contexts appeared only once.

Table 6: Characteristics of included sources in the cleaned extraction dataset.

Study ID	Year	Design	Sport/context	Level/population	Breathing category
Durand2004	2004	Observational cross-sectional	Endurance	Competitive	Other/unclear
Spiering2004	2004	Experimental, other	Various	Competitive	Other/unclear
Lindholm2005	2005	Experimental, other	Freediving	Elite	Other/unclear
Malakhov2014	2014	Experimental, other	Various	Competitive	Hyperventilation
Sakamoto2014	2014	Experimental, other	Various	Competitive	Hyperventilation
Sakamoto2015	2015	Experimental, other	Various	Competitive	Hyperventilation
Stavrou2015	2015	Experimental RCT	Swimming	Competitive	Hypoventilation
Hakked2017	2017	Experimental RCT	Swimming	Competitive	Pranayama/yogic
Sakamoto2018	2018	Experimental, other	Various	Competitive	Hyperventilation
Fernandez2019	2019	Experimental, other	Freediving	Elite	Other/unclear
Citherlet2021	2021	Experimental, other	Running	Recreational	Hyperventilation; Wim Hof
Taylor2021	2021	Experimental, other	Not sport-specific	Recreational	Paced breathing
Vaidya2021	2021	Experimental, other	Cricket	Competitive	Pranayama/yogic
Bahensky2021	2021	Experimental RCT	Running	Competitive	Pranayama/yogic
Marko2022	2022	Experimental RCT	Running	Elite	Hyperventilation; Wim Hof
You2023	2023	Experimental, other	Not sport-specific	Recreational	Paced breathing
Braham2024	2024	Experimental RCT	Soccer	Competitive	Hypoventilation
Li2024	2024	Experimental, other	Not sport-specific	Recreational	Paced breathing
Merlin2024	2024	Experimental RCT	Swimming	Elite	Paced breathing
Sikora2024	2024	Observational cross-sectional	Various	Competitive	Other/unclear
Woorons2024	2024	Experimental RCT	Judo	Elite	Hypoventilation

Study ID	Year	Design	Sport/context	Level/population	Breathing category
Deniz2025	2025	Experimental, other	Resistance training	Recreational	Breath holding; Valsalva
Fernandez2025	2025	Experimental, other	Freediving	Elite	Other/unclear
Fesseler2025	2025	Experimental, other	Running	Recreational	Hyperventilation
Fox2025	2025	Experimental RCT	Not sport-specific	Recreational	Hyperventilation; Wim Hof
Kasap2025	2025	Experimental, other	Cycling	Recreational	Paced breathing
Yilmaz2025	2025	Experimental, other	Team sports	Competitive	Hyperventilation
Chiron2026	2026	Experimental RCT	Track and field	Elite	Paced breathing
Devipriya2026	2026	Experimental before-after	Not sport-specific	Recreational	Pranayama/yogic
FernandezBarradas2026	2026	Experimental before-after	American football	Competitive	Paced breathing
Iskra2026	2026	Experimental, other	Not sport-specific	Recreational	Paced breathing
Katz2026	2026	Experimental, other	Resistance training	Recreational	Other/unclear
Raidl2026	2026	Experimental, other	Not sport-specific	Recreational	Paced breathing
Tomita2026	2026	Experimental, other	Skating	Competitive	Paced breathing

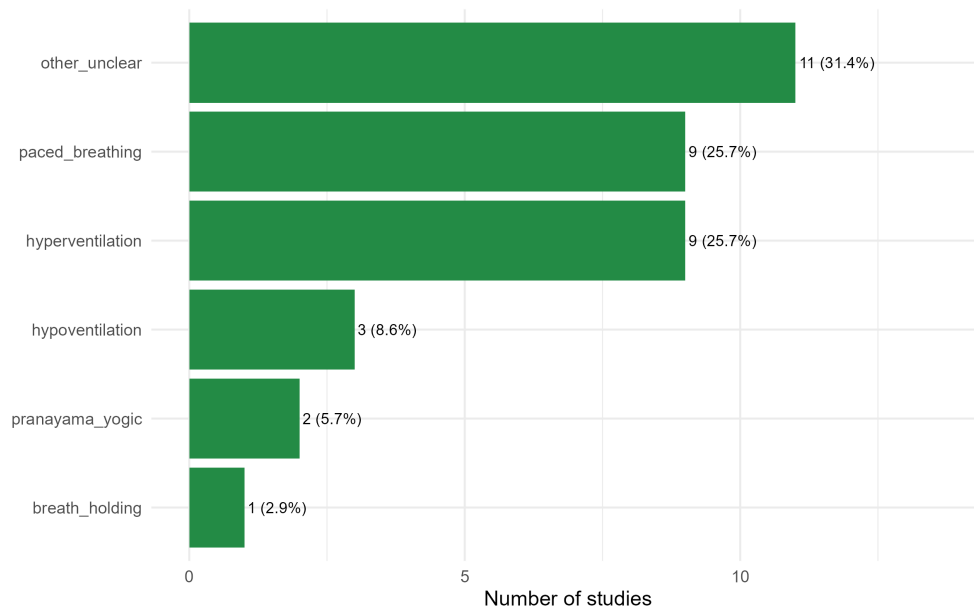


Figure 2: Distribution of breathing-training categories in the cleaned extraction.

3.3 Breathing-intervention categories and dose

The most frequent extracted categories were other/unclear breathing strategies, hyperventilation, and paced breathing. Hypoventilation, pranayama/yogic breathing, and breath holding or Valsalva-type strategies were less common. Dose reporting was uneven. Among parseable records, the median frequency was one session per week, and the median duration was one day, indicating that much of the literature concerns acute or short exposure protocols rather than extended training programs.

3.4 Publication trends, design, and sport distribution

The available evidence appears to be expanding, especially in the most recent extraction years. The trend should be interpreted cautiously because 2026 sources may represent early online records and because the exact search date is not available in the project files.

3.5 Outcome synthesis

Outcome measures were highly heterogeneous, covering pulmonary function, blood gases, sprint and power outcomes, reaction and movement time, heart-rate variability, perceived exertion, anxiety, recovery states, oxygen saturation, and sport-specific performance. Because most outcome domains were represented by few studies, the scoping synthesis emphasizes mapping rather than definitive effect estimation.

Exploratory quantitative synthesis was possible for two outcome clusters. Hyperventilation-related power outcomes showed a small pooled effect favoring the intervention ($g = 0.30$, 95% CI -0.00 to 0.60, $p = 0.052$, $I^2 = 0.0\%$). This estimate is compatible with a small beneficial effect but is borderline and should not be read as conclusive. Paced-breathing reaction outcomes were inconclusive ($g = -0.15$, 95% CI -0.45 to 0.15, $p = 0.335$, $I^2 = 57.9\%$). A hypoventilation repeated-sprint cluster had only one eligible effect and was not pooled.

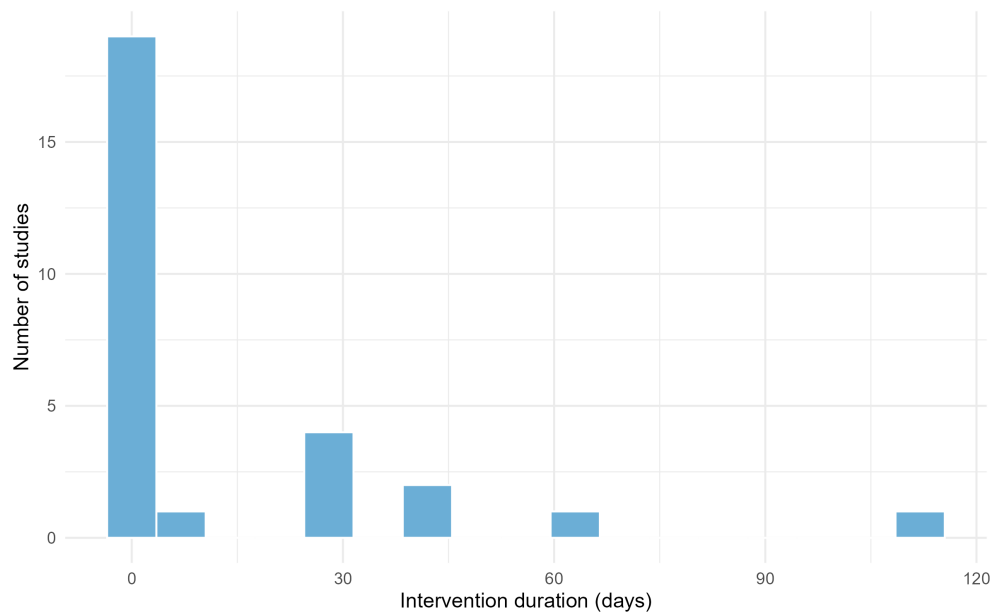


Figure 3: Distribution of intervention duration among parseable records.

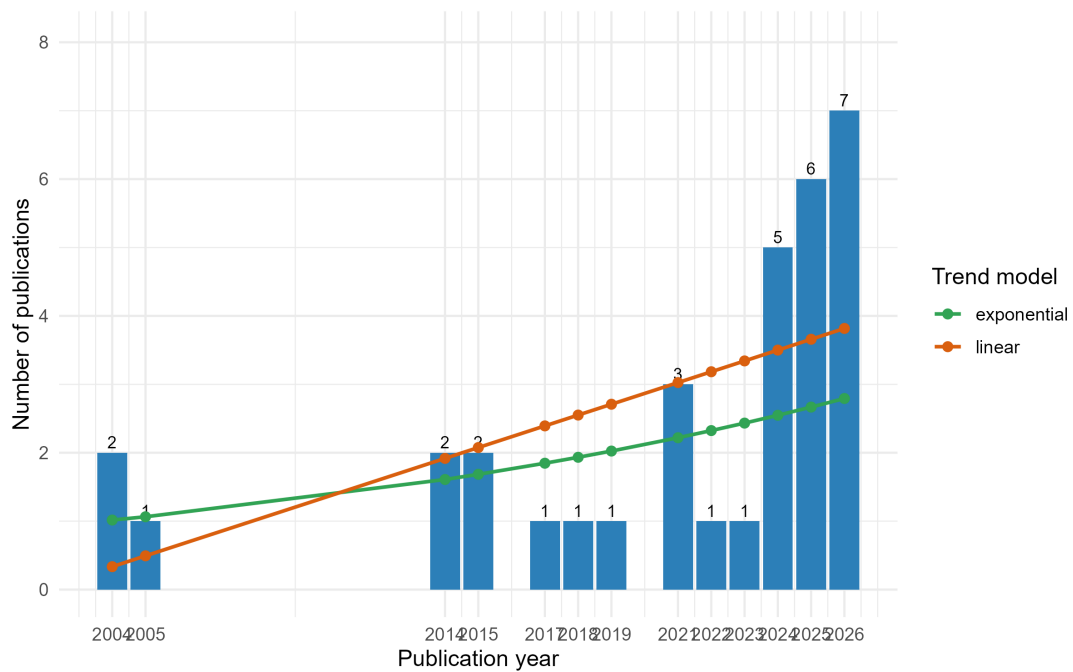


Figure 4: Annual publication counts and fitted exploratory trend lines.

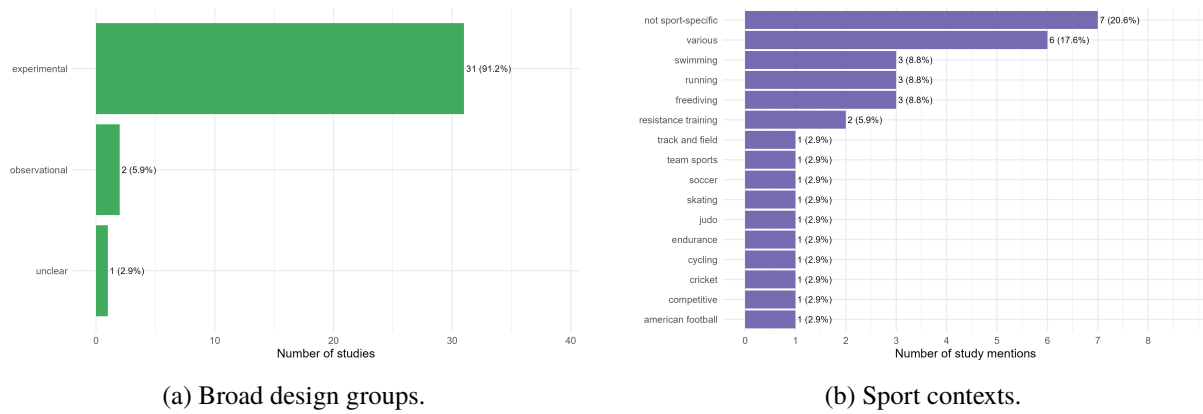


Figure 5: Distribution of study design and sport context.

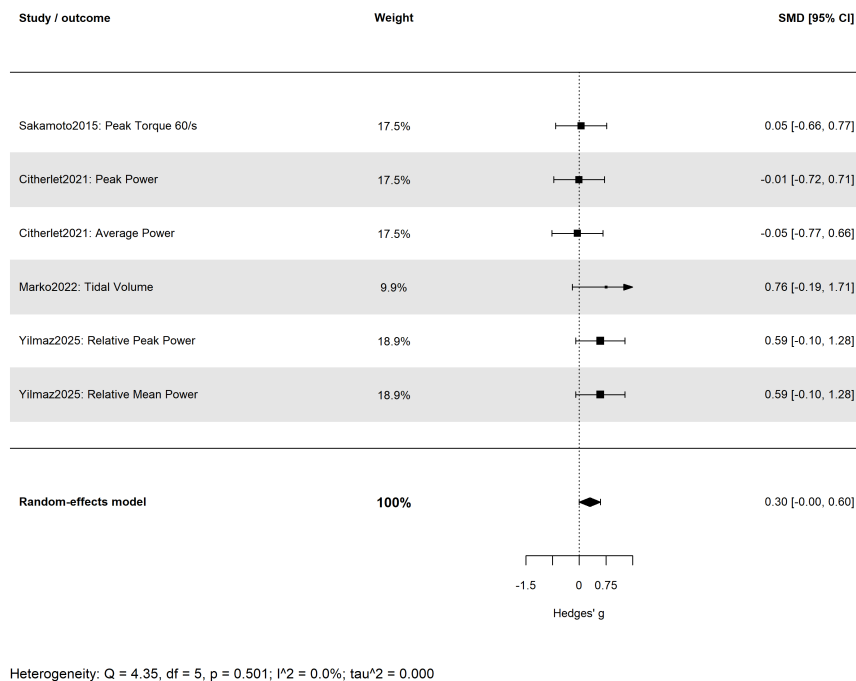


Figure 6: Exploratory forest plot for hyperventilation and power-related outcomes.

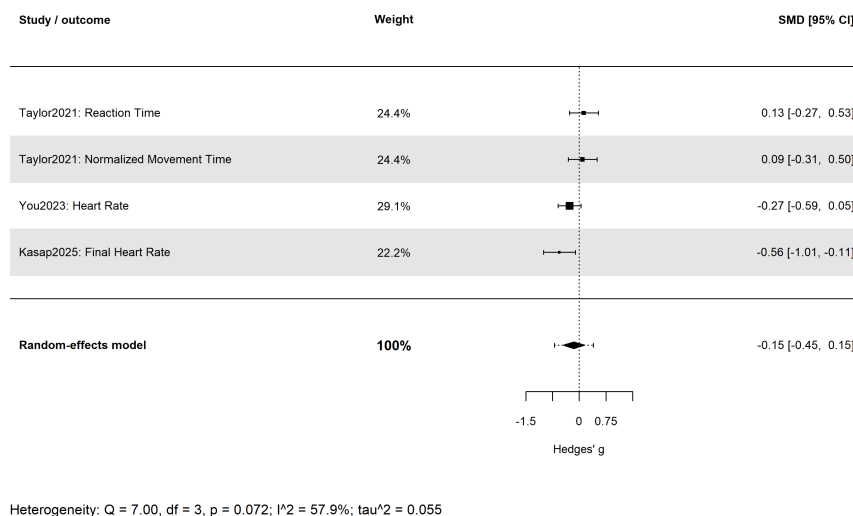


Figure 7: Exploratory forest plot for paced breathing and reaction-related outcomes.

3.6 Feasibility and safety

Feasibility reporting was sparse. The cleaned outputs contained no parseable adherence summaries and no parseable dropout summaries. Adverse-event extraction identified sensory symptoms associated with hyperventilation, including dizziness, heaviness, numbness, and tingling, and pulmonary edema in a freediving/hook-breathing context. These findings do not establish incidence rates because reporting was incomplete and inconsistent, but they indicate that safety reporting should be mandatory in future studies of respiratory maneuvers that alter carbon dioxide, oxygen saturation, intrathoracic pressure, or autonomic state.

4 Discussion

4.1 Summary of evidence

This scoping review mapped 34 sources on breathing-training and breath-regulation strategies in sport, exercise, and performance-relevant contexts. The evidence base is broad but fragmented. The included sources span several mechanistically different practices, including hyperventilation, hypoventilation, paced breathing, pranayama/yogic breathing, breath holding, Valsalva-type strategies, hook breathing, and Wim Hof-type breathing. Most studies were experimental, but the dominant pattern was small, acute, crossover, pilot, or repeated-measures work rather than large, preregistered, sport-specific trials.

The most practically relevant finding is not that one breathing method is clearly superior. Rather, the map shows that different breathing strategies are being used for different purposes and should be evaluated separately. Hyperventilation studies tend to target high-intensity power, sprint, blood-gas, or fatigue-related outcomes. Paced breathing is more often directed toward reaction time, recovery, heart-rate variability, stress, or anxiety. Hypoventilation and breath-holding studies are concentrated in repeated-sprint, swimming, judo, freediving, or oxygen-tolerance contexts. Pranayama/yogic studies often examine

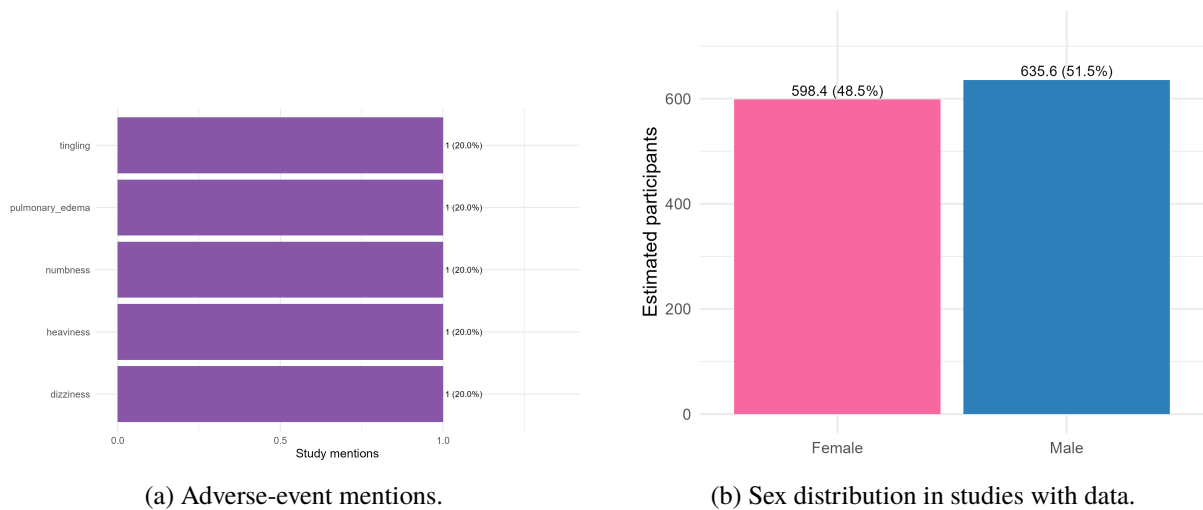


Figure 8: Safety and participant reporting summaries.

pulmonary, reaction-time, or broader performance-related outcomes.

The exploratory effect-size summaries are consistent with this interpretation. Hyperventilation showed a small, borderline pooled effect on power-related outcomes, but confidence intervals included a null effect. Paced breathing did not show a clear pooled benefit for reaction outcomes and showed moderate heterogeneity. These analyses are useful for identifying candidate domains for future systematic review, but they are not sufficient for definitive recommendations.

4.2 Implications for practice

Practitioners should avoid treating “breathwork” as a single intervention class. Acute hyperventilation, slow-paced breathing, hook breathing, hypoventilation, and pranayama differ in physiological targets, risk profile, and likely performance relevance. The mapped evidence suggests that some methods may be promising in narrowly matched contexts, such as hyperventilation before or between high-intensity bouts, paced breathing for psychophysiological regulation, and hypoventilation in repeated-sprint or aquatic contexts. However, practical use should remain conservative because many studies are small, short-term, and incompletely reported.

4.3 Implications for research

Future studies should define the breathing intervention precisely, including respiratory rate, depth, timing relative to exercise, session frequency, total exposure, supervision, and safety monitoring. Trials should select outcomes that match the hypothesized mechanism, report adherence and dropout, and preregister analysis plans. Sport-specific studies should distinguish elite, competitive, recreational, and non-athlete populations. The field would also benefit from a shared taxonomy separating ventilatory manipulation, autonomic regulation, respiratory-muscle practice, apnea/hypoventilation exposure, and meditative or yogic breathing practices.

4.4 Limitations

This manuscript was generated from the available AIK-SCR extraction and analysis files. The project folder did not contain a populated ‘paper/docs/introduction.md’, an exact PubMed search strategy, a search date, duplicate-removal details, or full-text exclusion reasons. Some bibliographic records in ‘articles.bib’ were difficult to reconcile with extraction IDs; Fernandez 2019, Fernandez 2025, and Devipriya 2026 were present in the extraction material but not reliably matched to a BibTeX entry by the available Rayyan export keys. One cleaned row lost the lowercase ‘bahensky2021’ identifier because the cleaning rule expected uppercase AuthorYear formatting. These issues should be reconciled before journal submission.

The effect-size analyses also have limitations. They rely on extracted group means and dispersions treated as standard deviations, combine only small clusters, and normalize effect directions algorithmically. As a result, the quantitative summaries should be interpreted as exploratory signals within a scoping review, not as conclusive evidence of efficacy.

5 Conclusions

The AIK-SCR evidence map shows a rapidly developing but heterogeneous literature on breathing strategies in sport and performance settings. The current evidence is strongest as a map of concepts, contexts, and gaps. Hyperventilation may have a small acute signal for power-related outcomes, while paced-breathing effects on reaction outcomes remain uncertain. Feasibility and safety reporting are major weaknesses across the corpus. Publication-ready future work should use clearer taxonomies, complete reporting, and sport-specific, adequately powered comparative designs before strong practical recommendations are made.

Declarations

Funding: No funding information was available in the project files.

Conflicts of interest: No conflicts of interest were recorded in the project files.

Data availability: The extracted data, analysis scripts, generated tables, and figures are available in the AIK-SCR project folders ‘Data’, ‘scripts’, and ‘outputs’.

Author contributions: Data extraction, cleaning, analysis, and manuscript drafting should be completed with named contributors before submission.

A PRISMA-ScR Checklist Crosswalk

Table 7: PRISMA-ScR reporting checklist crosswalk for this manuscript.

Item	Reporting element	Location in manuscript
Title	Identifies the report as a scoping review	Title page

Item	Reporting element	Location in manuscript
Abstract	Structured summary including background, objectives, eligibility, sources, charting, results, conclusions	Abstract
Rationale	Rationale for mapping the evidence	Introduction
Objectives	Review question and objectives	Introduction
Protocol and registration	Protocol status reported	Methods
Eligibility criteria	Population, concept, context, source characteristics	Methods
Information sources	PubMed and citation backtracking reported	Methods
Search	Search source count reported; exact string/date unavailable	Methods; limitations
Selection of sources	Screening stages and counts reported	Methods; Figure 1
Data charting process	Extraction, cleaning, and charting variables reported	Methods
Data items	Bibliometric, demographic, intervention, outcome, feasibility, and safety variables reported	Methods
Critical appraisal	Not performed as a formal quality appraisal; data completeness appraised descriptively	Methods; limitations
Synthesis	Narrative mapping plus exploratory effect-size summaries	Methods; Results
Selection results	Counts at each screening stage	Results; Figure 1
Characteristics	Included-source characteristics	Results; Table 6
Critical appraisal results	Data-quality issues summarized where relevant	Results; limitations
Individual-source results	Study-level characteristics and outcome clusters summarized	Results; appendices
Synthesis of results	Evidence map, plots, tables, and exploratory syntheses	Results
Summary of evidence	Main findings linked to objectives	Discussion
Limitations	Review and evidence-base limitations	Discussion
Conclusions	Interpretation and next steps	Conclusions
Funding	Funding statement placeholder included	Declarations

B Referenced Included Sources

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